

Machine Learning-Assisted Materials Design and Discovery of Low-Melting-Point Inorganic Oxides for Low-Temperature Cofired Ceramic Applications

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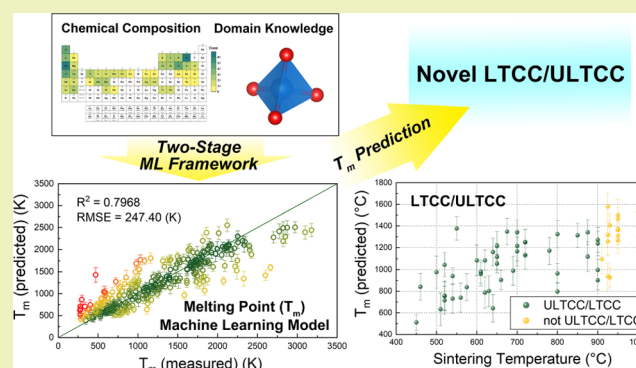
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ABSTRACT: The fabrication of low-temperature cofired ceramics (LTCCs) densified at a low sintering temperature (<900 °C) is energy-saving and environmentally friendly. However, finding novel LTCC materials by the trial-and-error method is time-consuming and costly. The LTCC materials often have low melting points, so it is feasible to discover high-performance LTCC materials out of the low-melting-point ceramics. A two-stage machine learning framework was adopted to establish the melting-point prediction model for inorganic oxides. Chemical compositions were used as features in stage 1 modeling; while in stage 2, more features were integrated according to domain knowledge to optimize the prediction model. Stage 2 model built by an artificial neural network algorithm shows the best performances with $R^2 = 0.7968$ and root-mean-square error = 247.4 (K). Three features, including formation energy per atom (fepa), theoretical density (d), and number of atoms (na), were extracted as the decisive characteristics of inorganic oxides. The melting point demonstrates positive correlations with the absolute value of fepa and d . The na acts as a “recessive gene” because its contribution is indirect but necessary. The physical relationships between features and the melting point were also discussed. Furthermore, the LTCC inorganic oxides often have melting points lower than 1400 °C statistically. This criterion was verified by the reported LTCC/ultra-LTCC materials. The melting points of materials in the prediction set consisting of ~3600 inorganic oxides were calculated by the ML model, and thus, the underlying LTCC materials could be screened out efficiently.

KEYWORDS: machine learning, melting point, inorganic oxide, quantitative structure–property relationships, ULTCC, LTCC



INTRODUCTION

Low-temperature cofired ceramic (LTCC) technology has extensive applications in the fabrication of passive components including capacitors, filters, couplers, balun, and so forth, which are the core components for wireless communication. By using the LTCC technology, ceramic substrates, metallic electrodes, and components can be formed as a monolithic structure at a low sintering temperature (S.T., <900 °C). Because of the low S.T., LTCC fabrication is not only energy-saving and environmentally friendly but also beneficial to extend the service life of the equipment (such as furnaces).^{1,2} The LTCC materials are typically glass-ceramic systems, whose densification is achieved by liquid-phase sintering.^{3,4} The disordered arrangement of glass network structures can hardly avoid deteriorating the dielectric, mechanical, and thermal properties of LTCC materials, so crystalline additives with a low melting point are widely investigated. The sintering additives, including CuO, LiF, Bi₂O₃, B₂O₃, and V₂O₅, can effectively reduce the S.T. and maintain the comprehensive properties of the major phase.^{5–9} However, the extrinsic dielectric loss may drastically rise due to the secondary phases.

In the past 2 decades, the National Sustainable Development Strategy (NSDS) proposed by the United Nations has driven the rapid development of ultralow temperature cofired ceramics (ULTCCs). The ULTCC materials have an intrinsic low S.T. (<700 °C), so they are free of the adverse effects of sintering additives. The reported ULTCC material systems are mostly tungstates, molybdates, tellurates, vanadates, and borates with cations of Li, Na, K, Ag, Ba, and Bi.^{10–14} So far, the ULTCC materials have been found by the trial-and-error method in the scope of the promising systems, which is time-consuming and costly. To discover new LTCC/ULTCC materials, it is feasible to find ceramics with low melting points. Sintering densification is associated with mass transportation

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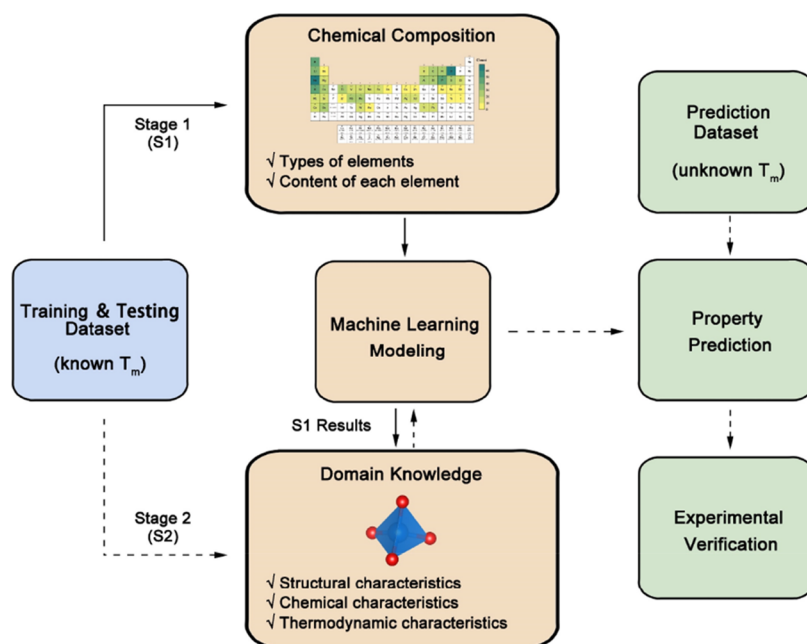


Figure 1. Flowchart of the two-stage ML procedures for the discovery of inorganic oxides with particular melting points.

and chemical bond formation. The particle morphology, size and gradation, density of green compact, pore distribution, sintering atmosphere, holding time, and other processing parameters have effects on the sintering process, which may influence the densification temperature. However, the decisive factor is the melting point of the material.¹⁵ The melting point is an intrinsic physicochemical property related to the crystal structure and elements, but it is hard to depict the quantitative relationship between them by theoretical analysis. Building a quantitative melting-point prediction model is significant to accelerate the discovery of LTCC/ULTCC materials.

Machine learning (ML) is a powerful tool of the fourth paradigm of science for seeking the quantitative structure–property relationship.^{16–21} Unlike the time-consuming and labor-intensive trial-and-error experiments, ML methods have an advantage to deal with complex problems in an efficient and sustainable way.^{22–25} In recent years, ML-assisted material informatic methods have made great progress in materials design and discovery.^{26–29} The ML modeling applied in melting point prediction has been reported but mostly for organic compounds.³⁰ The mature melting-point database of organic compounds and descriptor systems for organic molecular structures lay solid foundations for modeling.^{31–33} Karthikeyan et al. derived the first general melting-point prediction model for organic materials based on a large data set comprising 4173 compounds via the artificial neural network (ANN) method.³⁴ Tetko et al. developed a support vector machine melting-point prediction model using the data of several hundred thousand compounds extracted from published patents by text mining.³³ More recently, Mi et al. transferred the 3D chemical structures into line notations by the natural language processing approach to reach the best reported performances [$R^2 = 0.83$, root-mean-square error (RMSE) = 36.88] of the melting-point prediction ML model for organic materials until now.³⁵ However, the ML modeling for the melting point prediction of inorganic materials is still in an infant stage. Compared with the organic compounds built by C, H, O, N, S, P, and halogens mostly, the inorganic

materials are more complicated because of the diversified elements and chemical bonding. Besides, there is no generalized and standardized melting-point database for inorganic materials. Therefore, only a few research studies have reported the melting point model aiming at a certain category of ceramics, such as ultrahigh-temperature ceramics (UHTCs). The prediction model built by Qu et al. associated the melting point with element contents of ceramics, but it only covered eight most common elements for nonoxide UHTCs.³⁶ Besides, structure–property relationships were not involved because the model inputs are element types and contents only. Chemical and structural factors, including the chemical bond, lattice energy, molecular weight, and electronegativities of different atoms, were elaborated governing the melting point of inorganic materials.³⁰ The ML melting-point prediction model for octet AB binary crystals was first established based on 16 atomic features with a small data set.³⁷ Seko et al. adopted the results of density-functional theory (DFT) calculations as predictor variables for the model, which improved model performances and also enlarged the prediction range to single and binary compounds.³⁸ Pilania et al. introduced a new feature of the excess Born effective charge to the ML melting-point prediction model of octet compounds, and the predictability was improved even based on a small data set.³⁹ Thus, it can be seen that combining suitable elemental, chemical, and structural characteristics is expected to propose an accurate and general prediction model to discover low-melting-point inorganic compounds for LTCC applications via ML methods.

In this work, we adopted a two-stage ML framework to establish a melting-point prediction model for inorganic oxides. The features in stage 1 (S1) modeling were the contents of different elements, while chemical and structural features were integrated to optimize the model according to the domain knowledge (DK) in stage 2 (S2). Finally, we validated the prediction results by the inorganic oxides not included in the modeling data set. Several promising low-melting-point

material systems were found, and they are worthy of further experimental study.

METHODS

Figure 1 shows the two-stage ML strategy we employed to construct and verify the melting-point prediction model. Melting point data of inorganic oxides was collected from handbooks to build the data set for ML.^{40–42} Different categories of material characteristics were chosen as features at different stages. For S1 modeling, the features are the contents of 42 elements in a chemical formula. The 42 types of elements and their frequencies in the data set are shown in Figure S1. For S2 modeling, except for using the prediction results of the S1 model, we also introduced DK to enlarge the features, consisting of the structural, chemical, and thermodynamic characteristics of materials (Table 1). Four commonly used ML algorithms, including

Table 1. Features Introduced in S2 Modeling Based on DK

notation	feature	notation	feature
<i>m</i> (g/mol)	molecular mass	<i>blsd</i> (Å)	standard deviation of bond length
<i>ne</i>	number of elements	<i>blnpv</i> (Å ^{−3})	number of bonds per unit volume
<i>na</i>	number of atoms	<i>d</i> (g/cm ³)	theoretical density
<i>amm</i> (g/mol)	average value of atomic mass	<i>lsd</i> (Å)	standard deviation of length (lattice parameter)
<i>amsd</i> (g/mol)	standard deviation of atomic mass	<i>asd</i> (deg)	standard deviation of angle (lattice parameter)
<i>enm</i>	average value of Pauling electronegativity	<i>vm</i> (Å ³)	volume of primitive cell
<i>ensd</i>	standard deviation of Pauling electronegativity	<i>va</i> (Å ³)	volume of primitive cell per atom
<i>iem</i> (eV)	average value of the first ionization energy	<i>bg</i> (eV)	band gap
<i>iesd</i> (eV)	standard deviation of the first ionization energy	<i>epa</i> (eV)	final energy per atom
<i>blm</i> (Å)	average value of bond length	<i>fepa</i> (eV)	formation energy per atom

linear regression (*lr*), random forest (*rf*), support vector regression with linear kernel (*svr.lin*), and that with radial basis function kernel (*svr.rbf*) were applied for developing regression models in S1. Except for the four algorithms, two new algorithms, including gradient boosting decision tree (*gbdt*) and *ann*, were used in S2. The whole data set was split randomly into 70% as the training set and 30% as the testing set. Hyperparameters of different algorithms were optimized by the grid search method with 10-fold cross-validation for 30 repetitions. In S2 modeling, 20 features newly introduced were reduced by two-step dimension reduction, that is, “redundancy reduction” and “unimportance reduction”. By the method of exhaustion, 63 feature combinations were obtained for finding the optimal model and key features for melting point prediction. The optimal model was adopted to predict the melting points of inorganic oxides not included in the handbooks. Inorganic oxides with predicted low melting points were found to be promising in LTCC/ULTCC applications.

RESULTS AND DISCUSSION

Data Collection and Pretreatment. The accuracy of a ML model is tightly correlated with the quality of the data set. To ensure the reliability of data, three published handbooks were selected as the melting point data sources. We targeted inorganic oxides as the research subject because they attract the most attention in the field of LTCC and have a large amount of data in the selected handbooks. In S1 modeling,

only the contents (numbers) of 42 elements in a chemical formula were used. The features used in S2 modeling are basic characteristics of materials, which may have relationships with the melting point. These feature data were exported directly or calculated based on the exported data from the Royal Society of Chemistry (<https://www.rsc.org/periodic-table>) and the Materials Project (<https://materialsproject.org/>).^{43,44} More details are shown in Table S1.

Data pretreatment is necessary for a robust model. The melting point data from different handbooks may be different. Therefore, an average value is preferable if the data values are close to each other. However, we removed these data with great differences for improving the quality of data set. Finally, we obtained a data set of 302 inorganic oxides with more than 94% of materials having melting points below 2273.15 K (Figure S2).

S1 Modeling. The elemental composition is the most straightforward characteristic of a material, which may have complicated relationships with the melting points of inorganic oxides. For example, with the same element types, BaSiO₃ ($T_m = 1878.15$ K) and BaSiO₄ ($T_m = 1693.15$ K) have different melting points, while with the same universal chemical formula, BaZrO₃ ($T_m = 2773.15$ K) is far more refractory than BaSiO₃. The physical rationale behind the relations could be exemplified by knowledge about the crystal structure and bond energy. However, collecting the related data is time-consuming. In contrast, elemental compositions are easy to obtain. The prediction model based on elemental compositions would be useful in the rapid screening of materials with required melting points.

Four commonly used ML algorithms, including *lr*, *rf*, and support vector regression with a linear kernel (*svr.lin*) and with a radial basis function kernel (*svr.rbf*), were employed to construct melting-point prediction models based on the elemental compositions in S1 modeling. Hyperparameter optimization was realized by the grid search method and 10-fold cross-validation for 30 repetitions (see Figure S3). The optimized models built by different algorithms were compared based on four criteria (Figure 2), namely, the coefficient of determination (R^2) for correlation evaluation, RMSE for error assessment, average standard deviation (σ) for data dispersion appraisal, and deviation rate (δ) distribution for data skewness appraisal. To ensure the reliability, 1000 repeated trials were employed to calculate the average values of criteria. The criteria are given by eqs S1–S4.

In Figure 2, we confirm that the best algorithm is the *rf* method because the *rf* model has the highest R^2 (0.6762), the lowest RMSE (201.42), and an average standard deviation (61.9), although the δ distribution shows a relatively large positive skewness (the average value is larger than the median). The *lr* model is far inferior than the *rf* model, which means that the relationship between the elemental compositions and melting point does not conform to a simple linear relation. The performances of ML models on the testing set are visualized as Figure 3. The diagonal line represents the ideal prediction, and the color tone of the point demonstrates the absolute deviation rate. The *rf* model has the best predictability of the melting point, and all the prediction values are above 0 K. In contrast, some melting points predicted by the other three models are lower than absolute zero, which are physically meaningless. However, in the low-melting-point range ($T_m < 750$ K), large deviation of the *rf* model can still be observed. This may be difficult to eliminate due to the limitation of features. Features

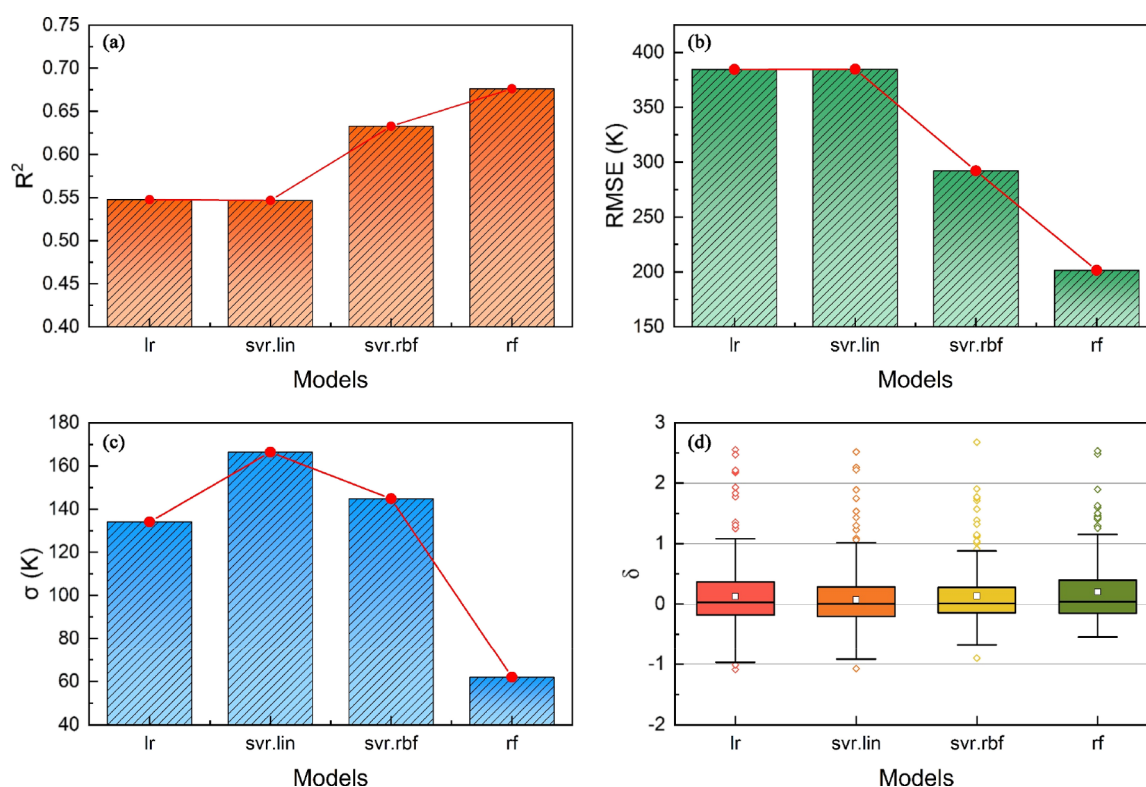


Figure 2. Comparison of predictability between different models in S1. (a) Coefficient of determination (R^2), (b) root-mean-square error, (c) average standard deviation (σ), and (d) deviation rate (δ) of the testing set.

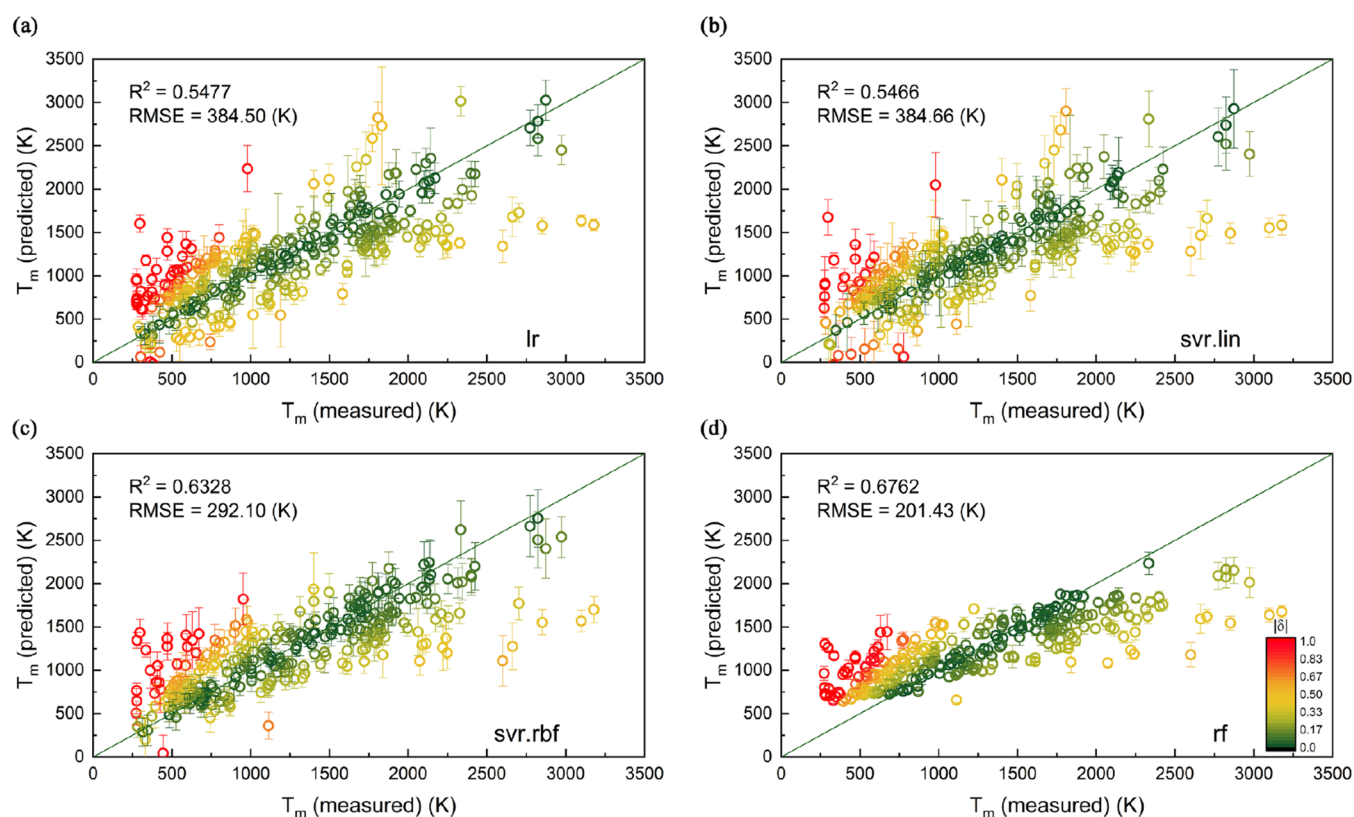


Figure 3. Performances of different models in the testing set in S1. (a) *lr*, (b) *svr.lin*, (c) *svr.rbf*, and (d) *rf*. The error bar represents the standard deviation (σ) of the predicted permittivity for 1000 repeated trials. The diagonal line indicates the ideal predicted results.

of elemental compositions cannot reflect the structural characteristics accurately. Since the melting point is closely

correlated with the crystal structure and chemical bonding, DK was introduced to overcome the shortcoming of the S1 model.

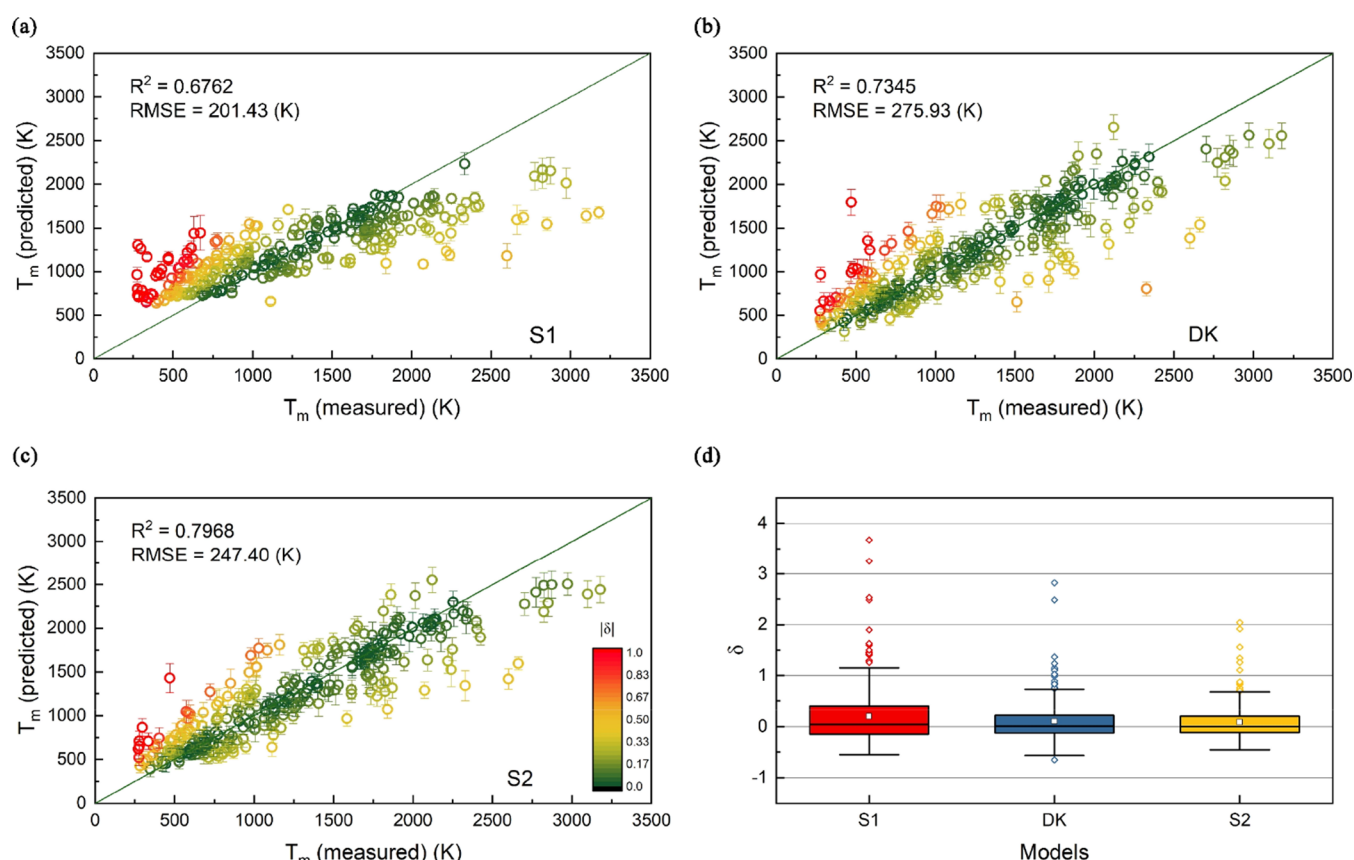


Figure 4. Performances of the optimized models (a) using elemental compositions only (S1), (b) using DK only, and (c) combining elemental compositions and DK (S2) in the testing set. The error bar represents the standard deviation of predicted permittivity for 1000 repeated trials. The diagonal line indicates the ideal predicted results. (d) Deviation rate distribution of the S1, DK, and S2 models.

S2 Modeling. The DK has been verified for its merits of improving the model performance in many works, especially for small data sets with not fully representative data, that is, data bias.^{45,46} Twenty features related to the composition, structure, and thermodynamics were introduced to improve the S2 model (Table 1). Considering the computational expense, model robustness, and interpretability, high-dimensional data is not applicable for modeling. Two-step dimension reductions, namely, “redundancy reduction” and “unimportance reduction”, were adopted before the subsequent procedures. The redundancy reduction was conducted by retaining only one feature among several highly correlated ones. The Pearson correlation coefficient (r) between each pair of features is shown in Figure S4. We selected the feature to be retained according to physical correlation with melting point and data accessibility. Sixteen decorrelated features with one another ($r < 0.8$) remained for the second step reduction. The unimportance reduction means discarding the less important features and extracting the most relevant one to the melting point. The *rf* algorithm was used for importance estimation, and the results are demonstrated in Figure S5. We retained the top six features, including the number of atoms (*na*), standard deviation of bond length (*blsd*), number of bonds per unit volume (*blnpv*), theoretical density (*d*), final energy per atom (*epa*), and formation energy per atom (*fepa*). Finally, high-dimensional features were reduced to the six key features for modeling.

The 6 remaining features after dimension reduction were combined exhaustively to generate 63 possible feature

combinations (Table S2). Different feature combinations and the predicted melting point of the S1 model were used as features in S2 modeling. Six ML algorithms, including *lr*, *rf*, support vector regression with a linear kernel (*svr.lin*) and with a radial basis function kernel (*svr.rbf*), *gbdt*, and *ann*, were employed to build the prediction S2 model. The grid search method optimized the hyperparameters by selecting the highest average R^2 of the 10-fold cross-validation for 30 repetitions (Figure S6a). Model performances and complexity were taken into account when we selected the best feature combination (Figure S6b,c). Comparison among different optimized models reveals that the *ann* model with features of *na*, *d*, and *fepa* has desirable outputs (Figure S7).

The integration of DK offers advantages in model predictability (Figure 4). For comparison, the optimized model using DK only was also shown. The highest R^2 , an acceptable RMSE, and improved skewness were obtained in the S2 model. Moreover, the number of samples with large prediction deviation in low (<750 K) and high (>2500 K) melting-point ranges was reduced, which means that the S2 model is applicable for T_m prediction in a wider temperature range. It should be noted that the DK model performs worse than the S2 model, but the outputs were better than the S1 model. The contributions from the DK outweigh those from the elemental compositions for melting point prediction, but the latter still contains irreplaceable and useful characteristics of materials for melting point prediction. For example, the cation–oxygen ratio dominates the coordination of cations, which contains the material characteristics related to

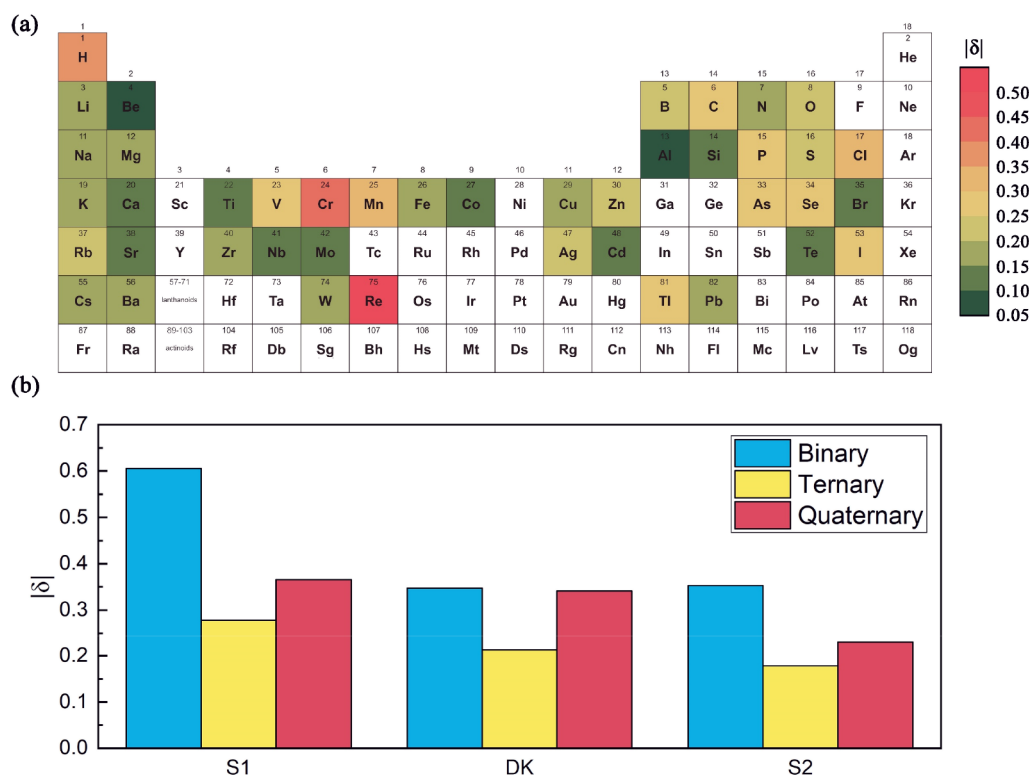


Figure 5. Absolute deviation rate of the predicted melting points (a) with different types of elements in the S2 model and (b) with different number of elements in S1, S2, and DK models.

symmetry, bonding, and networking. These characteristics have effects on the melting point, but they are difficult to quantify. Thus, the missing information is ascribed to the worse performance of the DK model than the S2 model.

The accuracies of the prediction models for the materials containing different elements are different (Figures S5a and S8). For the materials containing alkali metal (AM) or alkaline earth metals (AEMs), the prediction results are relatively accurate, while the average absolute deviation rates are higher than 0.2 for almost all materials containing nonmetals (NMs). The transition metals influence the model performances differently. The accuracy is high for Ti-, Zr-, Nb-, Co-, Cu-, and Cd-containing materials, but it is low for Cr, Re, Mn, and V. The difference in accuracy between the AM (or AEM) and NM is mainly derived from the overestimated results of some low-melting-point materials (Figure S9). The average absolute deviation rates of the low-melting-point ($T_m \leq 1273.15$ K) compounds are 0.2074 and 0.2684 for these two categories, respectively; while those of the high-melting-point ($T_m > 1273.15$ K) compounds are approximate (0.1139 and 0.1145, respectively). Besides, a larger proportion (64.03%) of low-melting-point compounds in the NM category than in the AM or AEM one (56.59%) magnifies the overall deviation. The explanation for the discrepancy of transition metals may lie in the amount and diversity of data. A large amount of data for model training is the premise for accurate prediction results. Owing to the enough data, the Ti-containing materials have high prediction accuracy, while the accuracy of Re-containing materials is low as a result of small data. However, small data may be enough for the modeling of compounds with the same cation–oxygen ratio. These compounds tend to have similar structural units, coordination, and chemical bonds. Therefore, they may share the same “gene” controlling the melting point.

A majority of Mo- and Zr-containing materials have Mo/O = 1:4 and Zr/O = 1:3, respectively. Thus, high prediction accuracies were obtained for them, although the amount of data was not large.

Most of the compounds in the data set are ternary, accounting for about 73%. The binary compounds take up about 23%, and the rest of the 4% materials are quaternary. Therefore, the unevenly distributed data may lead to data bias.⁴⁷ As shown in Figure 5b, the S2 model shows the best performances for the three types of compounds. Comparing S1 and DK models, the DK is of significance to the prediction accuracies for binary and ternary compounds. However, the accuracies of quaternary compounds are approximate. The prediction accuracy of quaternary compounds can be improved remarkably through combining the DK with the chemical compositions. The DK related to structural and chemical characteristics shows entanglement with the chemical compositions. Therefore, the relationships between the chemical compositions, DK, and melting point need further analysis.

Quantitative Feature–Property Relationship Analysis. Based on the prediction model, we can also extract the fundamental knowledge of the relationship between chemical and structural features and melting point of inorganic oxides. The optimal S2 model was based on the prediction results of the S1 model and three features, including f_{epa} , d , and na . The virtual space of inorganic oxides can be established in a 3D Cartesian coordinate system, which demonstrates the variation of the melting point upon changing the three features (Figures 6 and S10).

As shown in Figure 6, lower f_{epa} and d contribute to lower melting point, while the relation between na and melting point is not obvious. The f_{epa} and d are calculated using the first-

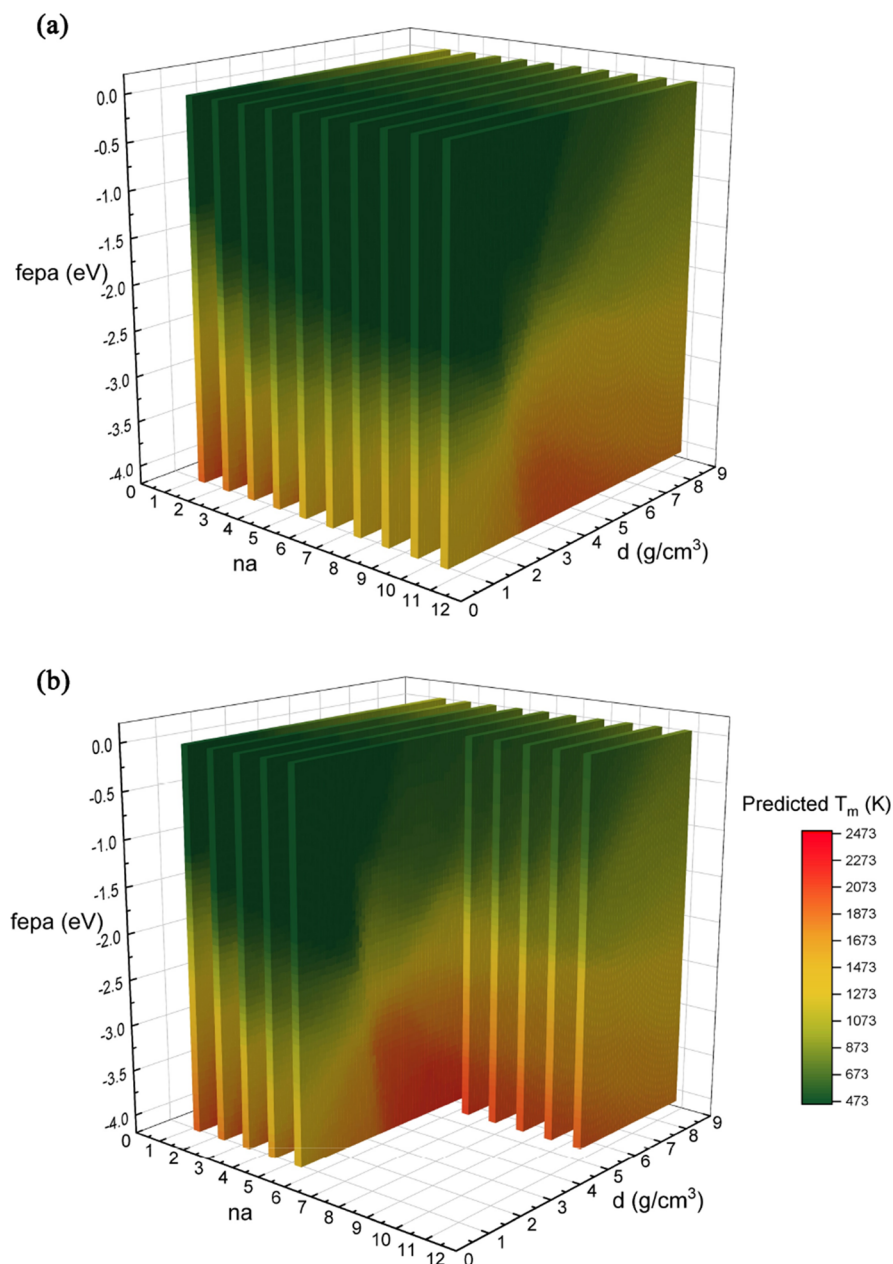


Figure 6. Distribution of (a) model-predicted melting points in the virtual space of materials and (b) cross section of (a) for a clear view of the center.

principles DFT, provided by the Materials Project database. The fepa is the formation energy of a compound normalized to per atom in a primitive cell, as given by eq 1

$$\text{fepa} = \frac{E - \sum \mu_i}{na} \quad (1)$$

where E is the total energy of the compound, μ_i is the chemical potential of the i atom, and na is the number of atoms in a unit cell.⁴⁸ Melting is a physical process, during which the intermolecular bonds are broken and the average kinetic energy of molecules is evaluated. The absolute value of the formation energy positively correlates with the bond energy. More negative formation energy per atom forms stable structures and strong intermolecular interactions, so the bond cleavage needs high external energy, which means that

the melting point of the material is high.^{30,49} Thus, the feature fepa plays a dominant role in melting-point prediction.

The theoretical density (d) mirrors the atomic packing in a crystal. The d value of a compound can be calculated according to eq 2

$$d = \frac{m}{N_A V_{pc}} \quad (2)$$

where m is the molecular mass, V_{pc} is the volume of a primitive cell of the compound, and N_A is the Avogadro constant. The molecular mass is governed by the mass and number of atoms comprised in a structure. The atomic mass is associated with the atomic radius. The atomic radius and the number of atoms determine the molecular surface area. Smaller molecular surface area weakens the steric repulsion, so the atoms can move more easily.^{50,51} On the other hand, the cell volume

reveals the average distance between atoms. In a large cell volume, the 3D network may be thermodynamically unstable if all the atoms are at great distances. A more possible situation is that the atoms form polyhedral units and link through their vertices. This 3D network is so called the “open-framework structure”.^{52,53} In the open-framework structures, large free space dilutes the intermolecular interactions. Not only the weakened steric effects and diluted intermolecular interactions reduce the melting point, but they also boost the mass transportation in the sintering process.

The material “genes”, a concept extrapolated from genetics, refer to the key factors controlling the target property of materials.⁵⁴ Not like the two features *fepa* and *d* acting as “dominant genes”, the *na* plays a role as a “recessive gene” for the melting point. The number of atoms (*na*) is correlated with the number of chemical bonds, but the contribution from *na* to the melting point seems small and unclear. However, the *na* is necessary for the optimal S2 model. The model without the feature *na* has a lower R^2 (0.7491) and higher RMSE (270.26), which is inferior to the optimal S2 model ($R^2 = 0.7968$ and RMSE = 247.40). The improvement implies that it is not suitable to analyze the relation between each feature and melting point separately. As per the aforementioned discussion, the *na* exists in the definition of the *fepa*, and it also affects the *d* value indirectly. The multifactorial interactions have effects on the property, and the relations may be too complicate to depict (Figure S11). Therefore, further analysis of the multifactorial interaction is challenging in theoretical perspectives, and the mathematical formula is nearly impossible to write out. With the assistance of the ML method, we enriched the human knowledge that the number of atoms (*na*) is a “recessive gene” controlling the melting point of inorganic oxides.

Model Verification and Prediction. Low-melting-point inorganic oxides not involved in the two-stage modeling were collected from published literature (Table S3). The new data was used to form the verification data set for evaluating the generalization capability of the S2 model. As shown in Figure 7, the low RMSE (109.70 K) verified the good predictability of these new materials. Therefore, the model is reliable for melting-point prediction of inorganic oxides.

As discussed above, it is reasonable to discover novel LTCC/ULTCC materials from the low-melting-point inorganic oxides. In order to depict the statistical relation between the S.T. and melting point (T_m), data of about 30

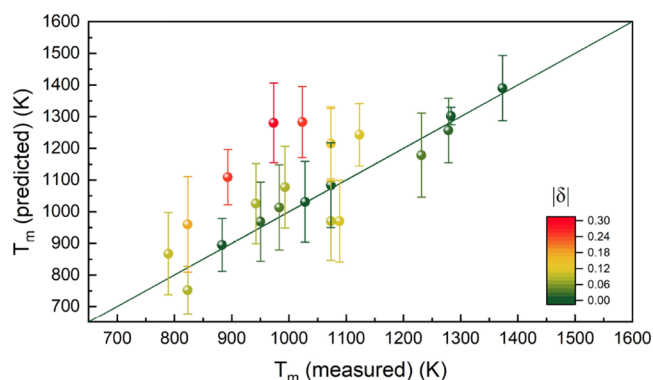


Figure 7. Performances of the ML model on the verification set. The error bar represents the standard deviation of predicted permittivity for 1000 repeated trials.

oxide ceramics were collected from the book written by Sebastian et al. (Table S4).⁵⁵ As shown in Figure 8a, the S.T.

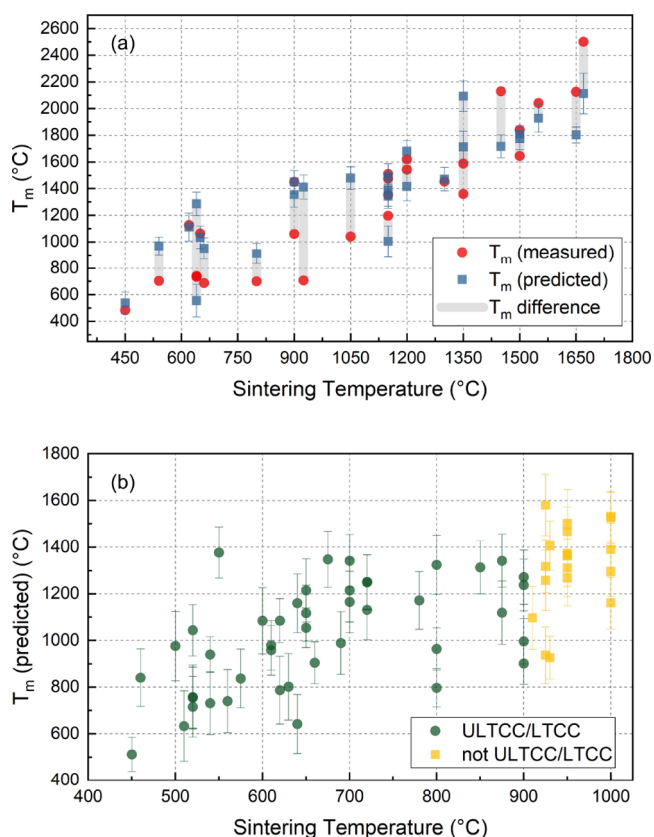


Figure 8. (a) Relation between the melting point and S.T. of some inorganic oxides with reported measured melting points. (b) Relation between the predicted melting point and S.T. of some single-phase ceramics.

of a ceramic has positive correlation with its melting point statistically. Almost all melting points are higher than the S.T. A majority of the ceramics with a S.T. lower than 900 °C have melting points lower than 1400 °C. Therefore, it can be used as a criterion to find the underlying LTCC/ULTCC materials. The measured melting points of many reported single-phase low-sintering-temperature ceramics cannot be found in the handbooks. However, the ML-predicted melting point can be calculated (Table S5). The similar relation between the predicted melting point and S.T. is shown in Figure 8b. Almost all ULTCC (S.T. < 700 °C) and LTCC (S.T. < 900 °C) materials have melting points lower than 1400 °C, which conforms with the criterion. Besides, some ceramics with a reported S.T. > 900 °C were predicted to have low melting points, such as $\text{Nb}_2\text{Cu}_3\text{O}_8$ (S.T. = 910 °C and predicted T_m = 1369.3 °C) and Li_2MnO_3 (S.T. = 930 °C and predicted T_m = 1199.2 °C). The S.T.s. of these ceramics can be possibly reduced by the optimization of fabrication processes so that they may satisfy the S.T. requirement of LTCC applications.

By using the criterion, the underlying LTCC/ULTCC materials can be discovered from the prediction data set efficiently. A prediction data set containing ~3600 inorganic oxides with a known chemical formula and the three features was established (see the Supporting Information). The melting points of the materials in prediction set have not been reported. By ML model prediction, several new inorganic

oxides with the predicted low melting point were found, including molybdate (CuMo_2O_7 , ZnMo_2O_7 , and Li_4MoO_5), tungstate (Cu_3WO_6 , $\text{Li}_6\text{W}_2\text{O}_9$, and KLiWO_4), borate (LiCoBO_3 , $\text{Co}_2\text{B}_2\text{O}_5$, and $\text{Na}_3\text{B}_7\text{O}_{12}$). These materials are worthy of further investigating to synthesize, measure their melting points, and be applied as LTCC/ULTCC materials.

CONCLUSIONS

A two-stage ML framework was adopted to establish a melting-point prediction model for inorganic oxides. The contents of different elements were used as features in S1 modeling, while the DK was integrated to optimize the model in S2. The *rf* model has the best predictability ($R^2 = 0.6762$ and $\text{RMSE} = 201.43 \text{ K}$) in S1. The ANN model shows the highest performances ($R^2 = 0.7968$ and $\text{RMSE} = 247.40 \text{ K}$) in S2. The introduction of three features, including formation energy per atom (*fepa*), theoretical density (*d*), and number of atoms (*na*), enhanced the performances of the ML model. Low absolute values of *fepa* and *d* contribute to low melting point directly. The *na* acts as a “recessive gene”, which means that its contribution is indirect but necessary. The S.T. has a positive correlation with the melting point statistically, and low-sintering-temperature ceramics (S.T. < 900 °C) often have melting points lower than 1400 °C. Several inorganic oxides without a reported S.T. were extracted from the prediction data set. Further experimental studies are needed to verify their LTCC/ULTCC applications.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.1c06983>.

Programming language and packages, important scores in unimportant reduction (S2 modeling), model tuning, and four criteria for model evaluation; data sources of features introduced in S2 modeling; combination of features in S2 modeling; measured and predicted melting point of inorganic oxides in the verification set; S.T., measured, and predicted melting point of some inorganic oxides; S.T. and predicted melting point of low-temperature sintering ceramics; frequency of elements in the data set; histogram of melting points of materials in the data set; optimization of hyperparameters in S1; map of the Pearson correlation matrix of 20 initial features; scores of importance of 16 redundancy-reduced features; model optimization of the ANN algorithm in S2; performances of optimized models using the S1 result and DK on the testing set; absolute deviation rate of the predicted melting points of inorganic oxides with different types of elements; deviation rate as a function of the melting point in the data set; and distributions of S1, S2, and DK model predicted and measured melting points in virtual space (PDF)

Training and testing sets (XLSX)

Verification and prediction sets (XLSX)

Raw data and codes for S1 and S2 modeling (ZIP)

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Sebastian, M. T.; Wang, H.; Jantunen, H. Low Temperature Co-Fired Ceramics with Ultra-Low Sintering Temperature: A Review. *Curr. Opin. Solid State Mater. Sci.* **2016**, *20*, 151–170.
- (2) Valant, M.; Suvorov, D. Processing and Dielectric Properties of Sillenite Compounds $\text{Bi}_{12}\text{MO}_{20-\delta}$ ($M = \text{Si, Ge, Ti, Pb, Mn, Bi}_{1/2}\text{P}_{1/2}$). *J. Am. Ceram. Soc.* **2001**, *84*, 2900–2904.
- (3) Zhou, J. Towards Rational Design of Low-Temperature Co-Fired Ceramic (LTCC) Materials. *J. Adv. Ceram.* **2012**, *1*, 89–99.
- (4) Sebastian, M. T.; Ubic, R.; Jantunen, H. Low-Loss Dielectric Ceramic Materials and Their Properties. *Int. Mater. Rev.* **2015**, *60*, 392–412.
- (5) Lai, Y.; Su, H.; Wang, G.; Tang, X.; Liang, X.; Huang, X.; Li, Y.; Zhang, H.; Ye, C.; Wang, X. R. Improved Microwave Dielectric Properties of $\text{CaMgSi}_2\text{O}_6$ Ceramics through CuO Doping. *J. Alloys Compd.* **2019**, *772*, 40–48.
- (6) Tao, B.; Wang, W.; Liu, H.; Du, T.; Wu, H.; Xing, C.; Wang, D.; Zhang, Y. Low-Temperature Sintering LiF -doped $\text{Li}_4\text{Mg}_3[\text{Ti}_{0.6}(\text{Mg}_{1/3}\text{Nb}_{2/3})_{0.4}]_2\text{O}_9$ Microwave Dielectric Ceramics for LTCC Applications. *Ceram. Int.* **2021**, *47*, 2584–2590.
- (7) Zhang, Q.-L.; Yang, H.; Tong, J.-X. Low-Temperature Firing and Microwave Dielectric Properties of MgTiO_3 Ceramics with Bi_2O_3 – V_2O_5 . *Mater. Lett.* **2006**, *60*, 1188–1191.
- (8) Dang, M.; Lin, H.; Yao, X.; Ren, H.; Xie, T.; Peng, H.; Tan, Z. Effects of B_2O_3 and MgO on the Microwave Dielectric Properties of MgTa_2O_6 Ceramics. *Ceram. Int.* **2019**, *45*, 24244–24247.
- (9) Huang, C.-L.; Chen, Y.-C. Influence of V_2O_5 Additions to NdAlO_3 Ceramics on Sintering Temperature and Microwave Dielectric Properties. *J. Eur. Ceram. Soc.* **2003**, *23*, 167–173.
- (10) Sasidharanpillai, A.; Kim, C. H.; Lee, C. H.; Sebastian, M. T.; Kim, H. T. Environmental Friendly Approach for the Development of Ultra-Low-Firing Li_2WO_4 Ceramic Tapes. *ACS Sustainable Chem. Eng.* **2018**, *6*, 6849–6855.

- (11) Hao, S. Z.; Zhou, D.; Hussain, F.; Su, J. Z.; Liu, W. F.; Wang, D. W.; Wang, Q. P.; Qi, Z. M. Novel Scheelite-Type $[\text{Ca}_{0.55}(\text{Nd}_{1-x}\text{Bi}_x)_{0.3}]\text{MoO}_4$ ($0.2 \leq x \leq 0.95$) Microwave Dielectric Ceramics with Low Sintering Temperature. *J. Am. Ceram. Soc.* **2020**, *103*, 7259–7266.
- (12) Wang, S.-F.; Hsu, Y.-F.; Wang, Y.-R.; Sung, C.-C. Ultra-Low-Fire $\text{Zn}_2\text{Te}_3\text{O}_8$ - TiTe_3O_8 Ceramic Composites. *J. Am. Ceram. Soc.* **2011**, *94*, 812–816.
- (13) Valant, M.; Popović, J.; Mihelj, M. V.; Burazer, S.; Altomare, A.; Moliterni, A. Oxide Crystal Structure with Square-Pyramidally Coordinated Vanadium for Integrated Electronics Manufactured at Ultralow Processing Temperatures. *ACS Sustainable Chem. Eng.* **2017**, *5*, S662–S668.
- (14) Wang, K.; Zhou, H.; Zhou, X.; Luan, X.; Hu, S.; Deng, J.; Li, S.; Liu, X. High Relative Permittivity $\text{Bi}_4\text{B}_{2+x}\text{O}_{9+3x/2}$ ($x = 1.5, 2, 2.5, 3$) Microwave Ceramics for ULTCC Technology. *Ceram. Int.* **2020**, *46*, 13841–13847.
- (15) Kuang, X.; Carotenuto, G.; Nicolais, L. A Review of Ceramic Sintering and Suggestions on Reducing Sintering Temperatures. *Adv. Perform. Mater.* **1997**, *4*, 257–274.
- (16) Chibani, S.; Coudert, F.-X.; Wang, C. Machine Learning Approaches for the Prediction of Materials Properties. *APL Mater.* **2020**, *8*, 080701.
- (17) Rajan, K. Materials Informatics. *Mater. Today* **2005**, *8*, 38–45.
- (18) Wei, J.; Chu, X.; Sun, X. Y.; Xu, K.; Deng, H. X.; Chen, J.; Wei, Z.; Lei, M. Machine Learning in Materials Science. *InfoMat* **2019**, *1*, 338–358.
- (19) Mahmood, A.; Wang, J.-L. Machine Learning for High Performance Organic Solar Cells: Current Scenario and Future Prospects. *Energy Environ. Sci.* **2021**, *14*, 90–105.
- (20) Jia, Y.; Hou, X.; Wang, Z.; Hu, X. Machine Learning Boosts the Design and Discovery of Nanomaterials. *ACS Sustainable Chem. Eng.* **2021**, *9*, 6130–6147.
- (21) Qin, J.; Liu, Z.; Ma, M.; Li, Y. Machine Learning Approaches for Permittivity Prediction and Rational Design of Microwave Dielectric Ceramics. *J. Mater. Chem. A* **2021**, *9*, 1284–1293.
- (22) Mahmood, A.; Wang, J.-L. A Time and Resource Efficient Machine Learning Assisted Design of Non-Fullerene Small Molecule Acceptors for P3HT-Based Organic Solar Cells and Green Solvent Selection. *J. Mater. Chem. A* **2021**, *9*, 15684–15695.
- (23) Kumar, S.; Ignacz, G.; Szekely, G. Synthesis of Covalent Organic Frameworks Using Sustainable Solvents and Machine Learning. *Green Chem.* **2021**, *23*, 8932–8939.
- (24) Hu, J.; Kim, C.; Halasz, P.; Kim, J. F.; Kim, J.; Szekely, G. Artificial Intelligence for Performance Prediction of Organic Solvent Nanofiltration Membranes. *J. Membr. Sci.* **2021**, *619*, 118513.
- (25) Hardian, R.; Liang, Z.; Zhang, X.; Szekely, G. Artificial Intelligence: The Silver Bullet for Sustainable Materials Development. *Green Chem.* **2020**, *22*, 7521–7528.
- (26) Roy, A.; Babuska, T.; Krick, B.; Balasubramanian, G. Machine Learned Feature Identification for Predicting Phase and Young's Modulus of Low-, Medium- and High-Entropy Alloys. *Scr. Mater.* **2020**, *185*, 152–158.
- (27) Liu, Y.; Guo, B.; Zou, X.; Li, Y.; Shi, S. Machine Learning Assisted Materials Design and Discovery for Rechargeable Batteries. *Energy Storage Mater.* **2020**, *31*, 434–450.
- (28) Qin, J.; Liu, Z.; Ma, M.; Liu, F.; Qi, Z.-M.; Li, Y. Structure and Microwave Dielectric Properties of Gillespite-Type $\text{ACuSi}_4\text{O}_{10}$ ($A = \text{Ca}, \text{Sr}, \text{Ba}$) Ceramics and Quantitative Prediction of the $Q \times f$ Value via Machine Learning. *ACS Appl. Mater. Interfaces* **2021**, *13*, 17817–17826.
- (29) Ward, L.; Agrawal, A.; Choudhary, A.; Wolverton, C. A General-Purpose Machine Learning Framework for Predicting Properties of Inorganic Materials. *npj Comput. Mater.* **2016**, *2*, 16028.
- (30) Šima, J. Structure-related Melting and Boiling Points of Inorganic Compounds. *Found. Chem.* **2016**, *18*, 67–79.
- (31) Stepišnik, T.; Škrlić, B.; Wicker, J.; Kocev, D. A Comprehensive Comparison of Molecular Feature Representations for Use in Predictive Modeling. *Comput. Biol. Med.* **2021**, *130*, 104197.
- (32) Sushko, I.; Novotarskyi, S.; Körner, R.; Pandey, A. K.; Rupp, M.; Teetz, W.; Brandmaier, S.; Abdelaziz, A.; Prokopenko, V. V.; Tanchuk, V. Y.; Todeschini, R.; Varnek, A.; Marcou, G.; Ertl, P.; Potemkin, V.; Grishina, M.; Gasteiger, J.; Schwab, C.; Baskin, I.; Palyulin, V. A.; Radchenko, E. V.; Welsh, W. J.; Kholodovych, V.; Chekmarev, D.; Cherkasov, A.; Aires-de-Sousa, J.; Zhang, Q.-Y.; Bender, A.; Nigsch, F.; Patiny, L.; Williams, A.; Tkachenko, V.; Tetko, I. V. Online Chemical Modeling Environment (OCHEM): Web Platform for Data Storage, Model Development and Publishing of Chemical Information. *J. Comput.-Aided Mol. Des.* **2011**, *25*, 533–554.
- (33) Tetko, I. V.; Lowe, D.; Williams, A. J. The Development of Models to Predict Melting and Pyrolysis Point Data Associated with Several Hundred Thousand Compounds Mined from PATENTS. *J. Cheminf.* **2016**, *8*, 1–18.
- (34) Karthikeyan, M.; Glen, R. C.; Bender, A. General Melting Point Prediction Based on a Diverse Compound Data Set and Artificial Neural Networks. *J. Chem. Inf. Model.* **2005**, *45*, 581–590.
- (35) Mi, W.; Chen, H.; Zhu, D.; Zhang, T.; Qian, F. Melting Point Prediction of Organic Molecules by Deciphering the Chemical Structure into a Natural Language. *Chem. Commun.* **2021**, *57*, 2633–2636.
- (36) Qu, N.; Liu, Y.; Liao, M.; Lai, Z.; Zhou, F.; Cui, P.; Han, T.; Yang, D.; Zhu, J. Ultra-High Temperature Ceramics Melting Temperature Prediction via Machine Learning. *Ceram. Int.* **2019**, *45*, 18551–18555.
- (37) Saad, Y.; Gao, D.; Ngo, T.; Bobbitt, S.; Chelikowsky, J. R.; Andreoni, W. Data Mining for Materials: Computational Experiments with AB Compounds. *Phys. Rev. B* **2012**, *85*, 104104.
- (38) Seko, A.; Maekawa, T.; Tsuda, K.; Tanaka, I. Machine Learning with Systematic Density-Functional Theory Calculations: Application to Melting Temperatures of Single- and Binary-Component Solids. *Phys. Rev. B* **2014**, *89*, 054303.
- (39) Pilania, G.; Gubernatis, J. E.; Lookman, T. Structure Classification and Melting Temperature Prediction in Octet AB Solids via Machine Learning. *Phys. Rev. B* **2015**, *91*, 659–664.
- (40) Lide, D. R. *CRC Handbook of Chemistry and Physics a Ready-Reference Book of Chemical and Physical Data*, 88th ed.; CRC Press: Boca Raton, FL, 2007.
- (41) Dale, L. P. *Handbook of Inorganic Compounds*, 2nd ed.; CRC press: Boca Raton, FL, 2016.
- (42) James, G. S. *Lange's Handbook of Chemistry*, 17th ed.; McGraw-Hill Education: New York, NY, 2017.
- (43) Jain, A.; Ong, S. P.; Hautier, G.; Chen, W.; Richards, W. D.; Dacek, S.; Cholia, S.; Gunter, D.; Skinner, D.; Ceder, G.; Persson, K. A. Commentary: The Materials Project: A Materials Genome Approach to Accelerating Materials Innovation. *APL Mater.* **2013**, *1*, 011002.
- (44) Ong, S. P.; Cholia, S.; Jain, A.; Brafman, M.; Gunter, D.; Ceder, G.; Persson, K. A. The Materials Application Programming Interface (API): A Simple, Flexible and Efficient API for Materials Data Based on Representational State Transfer (REST) Principles. *Comput. Mater. Sci.* **2015**, *97*, 209–215.
- (45) Murdock, R. J.; Kauwe, S. K.; Wang, A. Y.-T.; Sparks, T. D. Is Domain Knowledge Necessary for Machine Learning Materials Properties? *Integr. Mater. Manuf. Innov.* **2020**, *9*, 221–227.
- (46) Xue, D.; Balachandran, P. V.; Yuan, R.; Hu, T.; Qian, X.; Dougherty, E. R.; Lookman, T. Accelerated Search for BaTiO_3 -based Piezoelectrics with Vertical Morphotropic Phase Boundary Using Bayesian Learning. *Proc. Natl. Acad. Sci. U.S.A.* **2016**, *113*, 13301–13306.
- (47) Ghandeharioun, A.; Eoff, B.; Jou, B.; Picard, R. Characterizing Sources of Uncertainty to Proxy Calibration and Disambiguate Annotator and Data Bias. *IEEE/CVF International Conference on Computer Vision (ICCV), Seoul, South Korea; IEEE Computer Society: Seoul, South Korea, 2019; pp 4202–4206.*
- (48) Emery, A. A.; Wolverton, C. High-Throughput DFT Calculations of Formation Energy, Stability and Oxygen Vacancy Formation Energy of ABO_3 Perovskites. *Sci. Data* **2017**, *4*, 170153.

(49) Jursic, B. S. High Level Ab Initio and Density Functional Theory Study of Bond Dissociation Energy and Enthalpy of Formation for Hypochloric and Hypobromic Acids. *J. Mol. Struct.: THEOCHEM* **1999**, 467, 173–179.

(50) Mebane, R. C.; Schanley, S. A.; Rybolt, T. R.; Bruce, C. D. The Correlation of Physical Properties of Organic Molecules with Computed Molecular Surface Areas. *J. Chem. Educ.* **1999**, 76, 688–693.

(51) Pérez-Casas, S.; Moreno-Esparza, R.; Costas, M.; Patterson, D. Effect of Steric Hindrance and π Electrons on Alcohol Self-Association. *J. Chem. Soc., Faraday Trans.* **1991**, 87, 1745–1750.

(52) Zheng, C.; Li, Y.; Yu, J. Database of Open-Framework Aluminophosphate Structures. *Sci. Data* **2020**, 7, 107.

(53) Natarajan, S.; Mandal, S. Open-Framework Structures of Transition-Metal Compounds. *Angew. Chem., Int. Ed. Engl.* **2008**, 47, 4798–4828.

(54) Rajan, K. Materials Informatics: The Materials “Gene” and Big Data. *Annu. Rev. Mater. Res.* **2015**, 45, 153–169.

(55) Sebastian, M. T.; Jantunen, H.; Uvic, R. *Microwave Materials and Applications 2V Set, I&II*, 1st ed.; John Wiley & Sons, Ltd: New Delhi, India, 2017; p 715–927.

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